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MT Joint Pre-Colloquium Session 2

From thesis idea to proposal: the art of developing a research question

Today's session

Thesis

Idea **Proposal**

Data science topics
and research questions

Different research questions, different goals

Description

- Describe policy-relevant phenomenon as precisely as possible
- Both latent (unobservable, e.g., opinion in text, public attitudes), and manifest (e.g., solar panels, income) outcomes of interest
- **Data science aspect:** innovative data collection and processing, measurement, labeling, modeling, benchmarking

Explanation

- Explain effect of X on Y, involving a policy-relevant phenomenon/relationship
- Focus on identification of causal effect, estimation of size
- **Data science aspect:** innovative research design, data collection and processing, econometric modeling

Prediction

- Predict policy-relevant Y from X
- Focus on predictive performance
- Predicting indicators, forecasting, …
- **Data science aspect:** innovative data collection and processing, feature engineering, predictive (e.g. ML) modeling

Building blocks of a data science project

Requirements for a data science master's thesis

1. A research gap giving rise to a research question
2. A hypothesis or idea how the RQ can be answered using data and data science
3. Availability of all relevant data needed for the analysis
   - Accessible free of charge (or very small costs)
   - If not accessible: plan how to collect, obtain, or annotate data within 1-2 months from proposal submission
   - Coverage and representativeness appropriate for task
   - Reasonable data quality and plan to clean and process the data
4. Data storage and computing infrastructure needs that match what you have available
5. Several methods that are promising to conduct the analysis and are ambitious but match your skillset (can be learned in a few months)

The thesis proposal step by step

Thesis proposal: the questions

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1. What is your **research topic**?
2. What is your **research question**?
3. Why would **you like to explore** this question?
4. What are the scholarly **debates** to which your thesis will relate?
5. What type of **research methods** are you planning to perform?
6. What type of **data collection strategy** are you planning to conduct?

See **Moodle**: **https://moodle.hertie-school.org/mod/book/view.php?id=90756&chapterid=2771**

1. What is your research topic?

- The field of studies your project potentially relates to
  - ✓ Be sure to select **“Data Science”,** plus additional substantive topics if relevant

- Describe your thesis topic in one paragraph (max 100 words) as specific as you can.
  - ✓ Specify the field(s) or subfield(s) your thesis would fit in
  - ✓ Directly refer to thesis topics that were suggested by supervisors in their supervision plans (or the first session) if interested
  - ✓ Name the dataset(s) you want to work with and describe briefly what methods you plan to explore in your analysis.

2. What is your research question?

- Write down your research question – please list your key question(s) of interest (max 75 words)
  - ✓ Must be a question that you set out to tackle with empirical evidence
  - ✓ Descriptive, explanatory, prediction questions – all legit! Not: normative questions
  - ✓ Be as specific as possible! You can also list subquestions
  - ✓ Consider some examples (also refer to last session):
    - ✓ What are suitable validation approaches and quality assessments of ML-inferred building attributes across Europe?
    - ✓ What resolution does satellite imagery need to have to detect electricity infrastructure?
    - ✓ (How) can we predict roof type (flat vs non-flat) from urban form across several European countries?
    - ✓ What determined people's willingness to adopt COVID-19 contact tracing apps?
    - ✓ How does the public think about prioritization trade-offs regarding energy supply?
    - ✓ How can we quantify popularity of political elites at a global scale?

3. Why would you like to explore this question?

- Briefly elaborate on the motivation behind your question. What makes your research question interesting and relevant? What is the key 'puzzle' that you aim to uncover? (max 100 words)
  - ✓ Describe why your RQ is interesting from a research perspective
  - ✓ Describe the **method(s)** that you propose to use in more detail
  - ✓ Personal experiences are legit motivators, too! But keep in mind that you’re not supposed to take action with your thesis. Instead, the aim is to address a research question and generate evidence/knowledge

4. What are the debates to which your thesis will relate?

- Briefly describe or list academic or policy debates that your research project relates to (max 100 words)
  - ✓ Public/policy debates can both be motivators of your work and be ultimately informed by it
  - ✓ For us, it'd be informative to see what research literature (papers, books, etc.) you have read that you think your thesis would speak to
  - ✓ You don't have to provide a literature review at this point but getting started is very helpful

5. What type of research methodology are you planning to perform?

17. **Research Methods I:** At this stage, what type of research methods do you expect to lean toward or can you imagine working with? You can change your mind afterwards, but your answer will be helpful to match you adequately with a supervisor. *

- ◯ Quantitative Methods (Large-n: Obtaining extensive, generalisable knowledge)
- ◯ Qualitative Methods (Small-n: Obtaining intensive, in-depth knowledge)

18. **Research Methods II:** Please elaborate on your previous answer (optional; 100 words max.).

Select Quantitiative Methods

- Natural language processing
- Computer vision
- Forecasting
- (Online) experiment
- Survey data analysis
- Meta-analysis
- Machine learning methods
- …

Be as precise as possible!

6. What type of data collection strategy are you planning to conduct?

19. **Data Collection Strategy I:** Do you plan to collect new/original data for your thesis or do you plan to use secondary data (i.e.,already existing data)? *

- ◯ I plan to collect my own data
- ◯ I plan to use secondary data.
- ◯ Both
- ◯ Not sure yet

20. **Data Collection Strategy II:** Please elaborate on your data collection strategy (optional). Remember that your answer is only tentative at this point (100 words max.).

Name the dataset(s) you want to work with (**with source!**)

- Satellite image analysis
- Original survey data collection
- Online experiment
- Automated web data collection
- ...

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Data and methods open text field at the end

Methods

- Natural language processing
- Computer vision
- Forecasting
- (Online) experiment
- Survey data analysis
- Meta-analysis
- Machine learning methods
- ...

Data collection strategy

- Satellite image analysis
- Original survey data collection
- Online experiment
- Automated web data collection
- ...

Be as precise as possible!

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Thesis proposal: the questions

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Abstract describing your planned work (important!)
Research field, 1-sentence context, dataset, methods

Be precise!

Policy or methodological relevance
Why do you think this is an important research question and what methods do you propose?

Academic literature
How does your project relate to and extend existing literature?

List the method(s) you plan to work with

Name the category of the data you have listed under Q1 and the type of data analysis you'll perform

See **Moodle**: **https://moodle.hertie-school.org/mod/book/view.php?id=90756&chapterid=2771**

Exercise: Pitch your own research idea in 1 minute!

Exercise

Pitch your own research idea in 1 minute!

- "The topic I want to do research on is…"
- "My research question, in a nutshell, is…"
- "To inform my research, I plan to collect data from…"
- "In terms of methods/the analytic approach, I plan to apply…"
- "I am still unsure about…"

Common pitfalls

Common pitfalls

1. You propose a topic but not a research question.
2. You propose a PhD thesis but not an MSc/MPP/MIA thesis.
3. You propose an interesting but intractable question.
4. You don‘t feel ownership of your project.

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Research questions – good and bad

1. Does peacekeeping work?
2. When do states adjust their economic policies in response to economic downturns?
3. What was the effect of the Comey letter on the outcome of the 2016 U.S. presidential election?
4. The future of Bitcoin
5. Can we predict carbon emissions at the municipal level?
6. Should negative campaigning be forbidden?
7. How can governments promote electric vehicles?
8. Will Russia stop the war against Ukraine?

Looking at the questions above, what are good and bad examples for RQs? Why? Discuss!

Research questions – good and bad

1. Does peacekeeping work? Too broad
2. When do states adjust their economic policies in response to economic downturns?
3. What was the effect of the Comey letter on the outcome of the 2016 U.S. presidential election?
4. The future of Bitcoin Too broad
5. Can we predict carbon emissions at the municipal level?
6. Should negative campaigning be forbidden? Normative
7. How can governments promote electric vehicles? Too broad
8. Will Russia stop the war against Ukraine? Cannot be answered

Looking at the questions above, what are good and bad examples for RQs? Why? Discuss!

Scope - You’re not getting a PhD (yet)

Common pitfalls

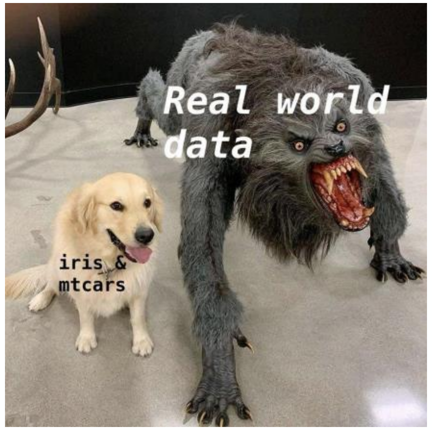
- Defining the research question too wide – probably you want to instead tackle a (sub)-sub-question of what you first had in mind
- Wanting the conditions to match your project – instead make the project match the conditions
- Trying to fit everything perfectly to your interests: method, topic area, social impact, etc. – MS thesis is just one project, not the project of your lifetime

Nevertheless, be ambitious! But know how to break things down if the project becomes too big.

The data

Common pitfalls

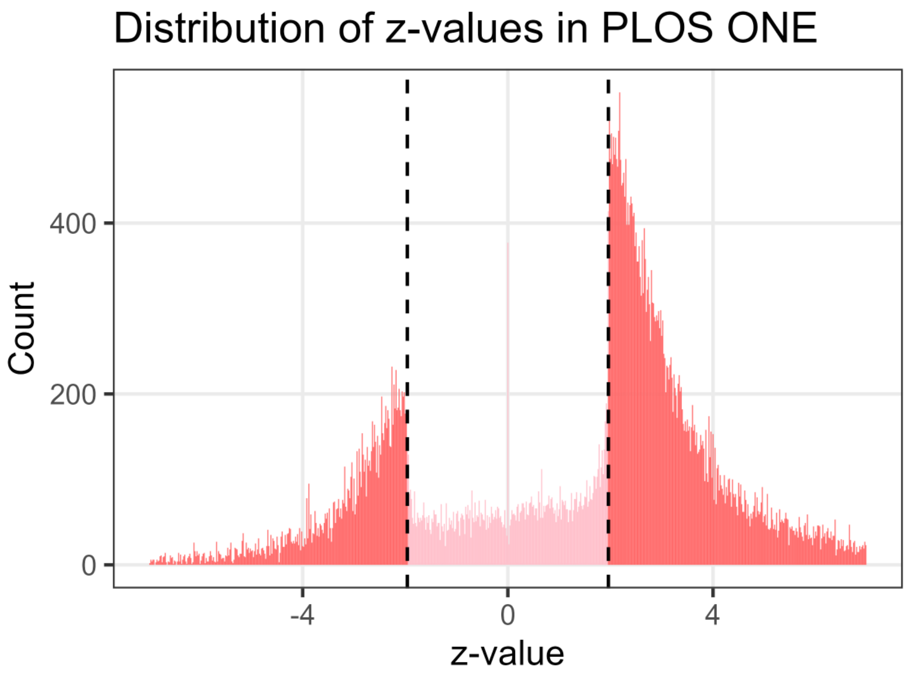
- Assuming data exist without identifying datasets (including "secondary" data, e.g. indicators you need to control for)
- Assuming data don't exist - sometimes they can be hard to find
- Underestimating data labeling
- Garbage in, garbage out (data quality and representativeness)
- Letting the data drive the research question (instead view it as iterative!)
- Working with the full data even if requiring large compute infrastructure
- Forgetting that 80% of the time are spent cleaning data when addressing real-world problems



The approach and methods

Common pitfalls

- If you have a hammer, everything looks like a nail – e.g., not everything requires deep learning
- Causation ≠ correlation
- Only considering one method – most research projects compare different methods to find the best one
- Validation: Not knowing what "best" actually means for your project
- Only expecting to find a meaningful result – a well-discussed nonresult can also make a great master's thesis



Practice partners

Tips for working with practice partners

Establishing your own partners

- Clarify expectations from all sides regarding IP, data release, etc.
- Provide agency to your thesis supervisor to shape the direction of the project if needed for you to pass the thesis
- Don't become a consultant

Working with partners suggested by faculty

- Indicate your interest early (= now) to the faculty and ask them what is the best approach to start getting in touch with the partner
- Manage the partnership yourself unless otherwise requested
- Remember that those in some cases are valuable professional contacts of the supervisor

Resources

Data

Example resources

Finding datasets:

- Look at related literature and the data they use
- Datasets listed in supervision plans
- Data repositories (e.g., Zenodo, Harvard Dataverse, …)
- Data journals (e.g., Nature Scientific Data)
- Government data platforms
- https://datasetsearch.research.google.com

Labeling data:

- If no labeled data are available, labeling may in some cases be an option
- Be prepared to spend a lot of time developing the annotation scheme, and a lot of repetitive work yourself (high quality) or use online services (lower quality)
- It is very ambitious to create a labeled dataset, do the analysis, and meaningfully draw policy conclusions from the data

Any questions?